DAIKIBO INDUSTRIALS

Manufacturing Downtime Analysis

Deloitte Australia Technology Virtual Experience

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Tool: Tableau

Dataset: Daikibo Telemetry Data (JSON)

160,704	4	9	103
Total Records	Factories	Machine Types	Unhealthy Events

1. Introduction

Daikibo Industrials is a global manufacturing company operating across four factories in Japan, Germany, and China. Each factory runs a set of manufacturing machines that continuously transmit telemetry data — including operational status and temperature readings — every 10 minutes.

This analysis was completed as part of the Deloitte Australia Technology Virtual Experience Program. The objective was to analyze machine telemetry data to identify which factories and machines experienced the most downtime, and deliver a Tableau dashboard enabling leadership to drill down from factory level to individual machine level in a single interactive view.

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2. Dataset Overview

Property	Detail
Dataset	Daikibo Telemetry Data
File Format	JSON
Total Records	160,704 Events
Analysis Tool	Tableau
Telemetry Frequency	Every 10 minutes per machine

2.1 Key Fields

Field	Description
deviceID	Unique machine identifier
deviceType	Type of manufacturing machine
timestamp	Event timestamp
location.country	Country of the factory
location.city	City of the factory
location.factory	Factory name
location.section	Section within the factory
data.status	Machine status: healthy or unhealthy

Field	Description
data.temperature	Machine temperature reading

2.2 Factories in Scope

Factory	Country	City	Total Records
daikibo-factory-meiyo	Japan	Tokyo	40,176
daikibo-factory-seiko	Japan	Tokyo	40,176
daikibo-berlin	Germany	Berlin	40,176
daikibo-shenzhen	China	Shenzhen	40,176

2.3 Machine Types Monitored

AirWrench, CNC, ConveyorBelt, Furnace, HeavyDutyDrill, LaserCutter, LaserWelder, MetalPress, SpotWelder

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3. Methodology

STEP 1 **Import JSON Data into Tableau**

The daikibo-telemetry-data.json file was imported into Tableau using a JSON File connection. All schema levels — including the root object, data, and location nested objects — were selected to expose all available fields for analysis.

STEP 2 **Create the Unhealthy Calculated Field**

A calculated field named Unhealthy was created to quantify downtime in minutes. Since telemetry messages are sent every 10 minutes, each unhealthy status reading represents 10 minutes of machine downtime:

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IF [Status] = "unhealthy" THEN 10 ELSE 0 END
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Summing this field per factory or device type gives the total downtime in minutes for any given grouping.

STEP 3 **Build Chart 1 — Downtime Per Factory**

A bar chart was built on Sheet 1 with Factory in the Columns field and SUM(Unhealthy) in the Rows field. This shows total downtime in minutes for each of the four factories.

STEP 4 **Build Chart 2 — Downtime Per Machine**

A second bar chart was built on Sheet 2 with Device Type in the Columns field and SUM(Unhealthy) in the Rows field. This shows total downtime across all factories broken down by machine type.

STEP 5 **Build the Interactive Dashboard**

Both charts were combined into a single Tableau Dashboard. Sheet 1 (Downtime Per Factory) was placed on top and Sheet 2 (Downtime Per Machine) below it. The Use as Filter option was enabled on Sheet 1 — allowing users to click any factory bar to filter Sheet 2 to show only the machine downtime for that specific factory.

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4. Status Distribution

160,601	103	99.94%	0.06%
Healthy Events	Unhealthy Events	Healthy Rate	Downtime Rate

While the overall unhealthy rate is very low at 0.06%, the 103 unhealthy events translate into significant cumulative downtime when measured in minutes — particularly when concentrated in specific factories and machine types.

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5. Key Findings

5.1 Downtime Per Factory

Factory	Country	Downtime (Mins)	Rank
daikibo-factory-seiko	Japan	480 mins	#1 — Most Downtime
daikibo-shenzhen	China	420 mins	#2
daikibo-factory-meiyo	Japan	110 mins	#3
daikibo-berlin	Germany	20 mins	#4 — Least Downtime

5.2 Downtime Per Machine Type

Device Type	Downtime (Mins)	Rank
LaserWelder	480 mins	#1 — Most Downtime
LaserCutter	430 mins	#2
HeavyDutyDrill	70 mins	#3
Furnace	20 mins	#4
CNC	10 mins	#5
SpotWelder	10 mins	#5
ConveyorBelt	10 mins	#5
MetalPress	0 mins	— No Downtime
AirWrench	0 mins	— No Downtime

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6. Key Insights

• daikibo-factory-seiko recorded the highest factory downtime at 480 minutes — making it the most critical location for maintenance intervention.

• daikibo-shenzhen follows at 420 minutes — together, the Seiko and Shenzhen factories account for 87% of all recorded downtime.

• daikibo-berlin is the best-performing factory with only 20 minutes of total downtime — its operational practices should be studied and replicated.

• LaserWelder is the most failure-prone machine type with 480 minutes of total downtime across all factories.

• LaserCutter is the second-highest at 430 minutes — both laser-based machines represent the highest operational risk.

• MetalPress and AirWrench recorded zero downtime — these machines are reliable and well-maintained across all facilities.

• The interactive dashboard allows leadership to click any factory and immediately see which specific machines failed there — supporting fast, targeted decision-making.

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7. Recommendations

1. Prioritize Seiko & Shenzhen

Immediate maintenance audits and inspection schedules should be deployed at daikibo-factory-seiko and daikibo-shenzhen — the two factories accounting for 900 combined minutes of downtime.

2. Investigate Laser-Based Equipment

LaserWelder and LaserCutter failures are the leading source of machine-level downtime. A root cause analysis should be conducted to determine whether this is a maintenance, calibration, or equipment quality issue.

3. Apply Berlin's Best Practices

daikibo-berlin's near-zero downtime suggests superior maintenance practices. These should be documented and replicated across all other facilities.

4. Implement Predictive Maintenance

Given that telemetry data is already being collected every 10 minutes, a predictive maintenance model could be built to flag machines approaching unhealthy status before failure occurs.

5. Continue Dashboard Monitoring

The Tableau dashboard should be refreshed regularly as new telemetry data is generated — enabling leadership to track downtime trends over time.

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8. Expected Outcomes

Recommendation	Expected Outcome
Maintenance audits at Seiko & Shenzhen	Reduction in factory-level downtime at the two highest-risk locations, improving overall production continuity.
LaserWelder & LaserCutter investigation	Resolution of root cause failures in the two highest-downtime machine types — potentially eliminating up to 910 minutes of combined downtime.
Apply Berlin best practices	Standardized maintenance protocols across all factories, reducing downtime variation between sites.
Predictive maintenance model	Early detection of machine failures before they occur — reducing reactive downtime and unplanned production stoppages.
Ongoing dashboard monitoring	Real-time visibility into factory and machine performance — enabling faster executive response to downtime events.

9. Conclusion

This analysis successfully transformed 160,704 raw telemetry events into a clear, actionable picture of manufacturing downtime across Daikibo Industrials' global factory network. By creating a calculated downtime measure and building an interactive two-level dashboard in Tableau, leadership can now identify both which factories are underperforming and which specific machines are responsible — all from a single view.

The most urgent finding is the concentration of downtime in daikibo-factory-seiko and daikibo-shenzhen, driven primarily by LaserWelder and LaserCutter failures. Addressing these two issues alone would resolve the vast majority of the platform's current downtime exposure.

Sign-Off

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Program: Deloitte Australia Technology Virtual Experience

Tool: Tableau

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